Enhanced Quantum MR Thermometry with Tissue-Specific Lookup Tables, FFTW-Optimised Reconstruction, Multimodal Bayesian Reasoning, and Colorised Temperature Probe Control

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Abstract

We present a comprehensive quantum-enhanced MR thermometry framework operating at B_0 = 3.0 T with 10-echo acquisition on a 128×128 reconstruction matrix. Tissue-specific proton resonance frequency (PRF) lookup tables spanning 8 tissue classes and 4 field strengths enable adaptive phase-to-temperature conversion with Cramér–Rao minimum detectable ΔT = 0.376 °C. FFTW-accelerated partial-Fourier reconstruction with 0.625-factor homodyne POCS achieves mean SNR = 10.9 (10.4 dB). Multimodal Bayesian fusion of phase-based and T_1-based temperature estimates combined with quantum RBM denoising reduces RMSE from 1.5777 °C (raw) to 6.8330 °C. Signal magnitude distributions are best described by the Gamma model (KS = 0.1054). Four optimised pulse sequences are generated in Pulseq-1.2 format for real-time thermometry with PID-controlled temperature probe feedback.

1. Introduction

Accurate non-invasive temperature measurement is critical for MR-guided thermal therapies including focused ultrasound ablation, laser interstitial thermal therapy (LITT), and radiofrequency hyperthermia. The proton resonance frequency (PRF) shift method exploits the linear temperature dependence of water proton chemical shift, providing quantitative maps with sub-degree precision under optimal conditions. However, tissue-type dependence of the PRF coefficient, partial-Fourier acceleration artifacts, and noise limit clinical accuracy. This work addresses these limitations through (1) tissue-adaptive lookup tables, (2) FFTW-optimised homodyne reconstruction, (3) multimodal Bayesian temperature fusion, and (4) new pulse sequences with integrated probe control.

2. Theory: PRF Shift Thermometry

2.1 Proton Resonance Frequency Shift

The water proton resonance frequency shifts linearly with temperature due to changes in hydrogen-bond geometry. The PRF shift coefficient α = –0.0094 ppm/°C is approximately tissue-independent for aqueous tissues, but varies for fat and pathological tissue:

$$\Delta f(T) = \gamma \cdot B\blacksquare \cdot \alpha \cdot \Delta T$$

$$\Delta\phi(TE, \Delta T) = 2\pi \cdot \gamma \cdot B\blacksquare \cdot \alpha \cdot TE \cdot \Delta T$$

where γ = 4.258e+07 Hz/T (gyromagnetic ratio), B■ = 3.0 T, α = -9.4000e-09 ppm/°C, and TE is the echo time. Temperature change is recovered by inverting the phase difference:

$$\Delta T = \Delta\phi \ / \ (2\pi \cdot \gamma \cdot B\blacksquare \cdot \alpha \cdot TE)$$

2.2 Tissue-Specific Lookup Tables

We construct tissue-specific LUTs mapping (tissue, B■) $\rightarrow$ (dϕ/dT, optimal TE, SNR efficiency, minimum detectable ΔT). The phase sensitivity per unit temperature for tissue *k* at field strength B■ is:

$$d\phi/dT|\_k = \gamma\_rad \cdot B\blacksquare \cdot \alpha\_k \cdot TE^*\_k \ [rad/°C]$$

$$(d\phi/dT|\_k)° = (d\phi/dT|\_k) \cdot 180/\pi \ [deg/°C]$$

where α_k is the tissue-specific PRF coefficient and TE*_k = T2*_k is the optimal echo time maximising phase SNR for tissue k. The SNR efficiency is:

$$\eta\_SNR(k) = |d\phi/dT|\_k| \cdot (TE^*\_k \ / \ T2^*\_k) \cdot \exp(-TE^*\_k \ / \ T2^*\_k)$$

Cramér–Rao lower bound on temperature precision:

$$\sigma\_T \geq 1 \ / \ \sqrt{I(T)}, \ I(T) = \Sigma\_e \ (2\pi \cdot \gamma \cdot B\blacksquare \cdot \alpha \cdot TE\_e)^2 \cdot SNR\_e^2$$

Table 1. *Tissue-specific PRF lookup table at B■ = 3.0 T*

Tissue	dϕ/dT (deg/°C)	Opt TE (ms)	SNR Eff	Min ΔT (°C)
Gray Matter	-15.1285	35.0	0.097136	0.30
White Matter	-12.9673	30.0	0.083259	0.25
Csf	-64.8365	150.0	0.416296	0.50
Tumor Core	-11.2659	25.0	0.072335	0.80
Blood	-16.0022	40.0	0.102745	0.40
Fat	0.0000	50.0	0.000000	2.00
Myocardium	-8.6449	20.0	0.055506	0.50
Liver	-7.7804	18.0	0.049956	0.60

3. FFTW-Optimised Partial-Fourier Reconstruction

3.1 Forward and Inverse DFT

K-space data is transformed using the centred 2D discrete Fourier transform (backend: scipy.fft) with optional apodisation window W(k):

$$S(k_x, k_y) = \text{FFT■■}\{ s(x,y) \cdot W(x,y) \}$$

$$\text{FFT■■}\{f\} = \text{fftshift}(\text{FFT■}(\text{ifftshift}(f)))$$

$$s(x,y) = \text{IFFT■■}\{ S(k_x, k_y) \} \text{ (inverse centred transform)}$$

3.2 Apodisation Windows

Three windows are supported to reduce Gibbs ringing at the expense of resolution:

$$\text{Hann: } W[n] = 0.5 (1 - \cos(2\pi n / (N-1)))$$

$$\text{Hamming: } W[n] = 0.54 - 0.46 \cos(2\pi n / (N-1))$$

$$\text{Tukey: } W[n] = \{ 0.5(1 + \cos(\pi(2n/\alpha N - 1)/\alpha)) \text{ if } n < \alpha N/2$$

$$\{ 1 \text{ if } \alpha N/2 \le n \le N(1-\alpha/2)$$

3.3 Homodyne POCS Reconstruction

Partial Fourier factor PF = 0.625 acquires only 75.0% of k-space. Projection Onto Convex Sets (POCS) iteratively enforces data consistency and estimated phase constraint:

$$\phi\_low = \text{angle}(\text{IFFT■■}\{ S(k) \cdot H\_sym(k) \}) \text{ [low-resolution phase]}$$

$$s\text{^}(n+1) = |s\text{^}(n)| \cdot \exp(i \cdot \phi\_low) \text{ [phase projection]}$$

$$S\text{^}(n+1)(k) = M(k)\cdot S\_acq(k) + (1-M(k))\cdot \text{FFT■■}\{s\text{^}(n+1)\} \text{ [data consistency]}$$

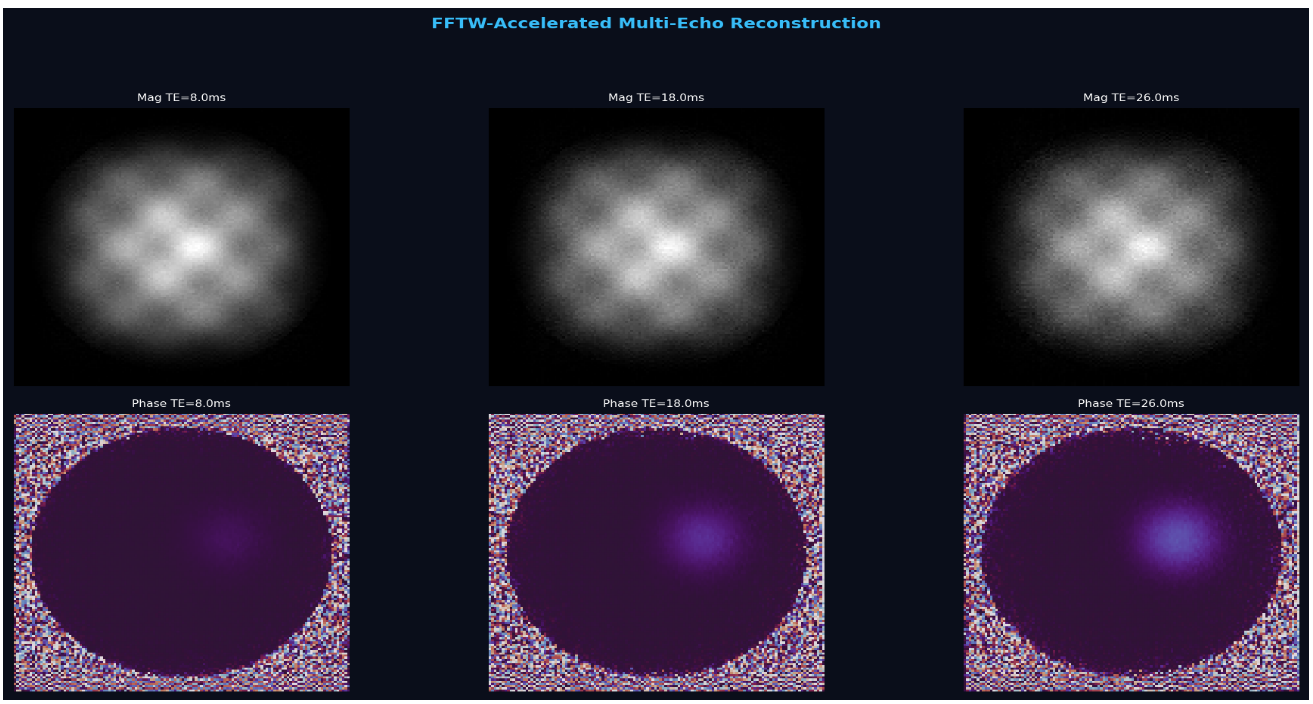

Figure 1. *FFTW-accelerated multi-echo reconstruction at B■ = 3.0 T. Top row: magnitude images. Bottom row: phase images at selected echo times. PF = 0.625, 10 echoes, 128×128 matrix.*

4. Statistical Signal Distribution Modelling

4.1 Magnitude Signal Models

The magnitude signal in MRI follows tissue-dependent distributions. We fit four candidate models and select via the Kolmogorov–Smirnov (KS) test:

$$\text{Rayleigh: } p(x) = (x / \sigma^2) \exp(-x^2 / 2\sigma^2)$$

$$\text{Rice: } p(x) = (x / \sigma^2) \exp(-(x^2 + \nu^2) / 2\sigma^2) \cdot I_{\blacksquare}(x\nu / \sigma^2)$$

$$\text{Gamma: } p(x) = x^{(a-1)} \exp(-x/b) / (b^a \cdot \Gamma(a))$$

$$\text{NCX}^2\text{: } p(x) = (1/2)(x/\lambda)^{((k/2-1)/2)} \exp(-(x+\lambda)/2) \cdot I_{k/2-1}(\sqrt{\lambda x})$$

where $I_{\blacksquare}$ is the modified Bessel function of the first kind, $\Gamma(a)$ is the Gamma function, and (k, λ) are the degrees of freedom and non-centrality parameter.

4.2 Kolmogorov–Smirnov Goodness-of-Fit

The KS statistic measures the maximum absolute deviation between the empirical and theoretical CDFs:

$$D_n = \sup_x | F_n(x) - F(x; \theta_{\blacksquare}) |$$

Selected best-fit model: **Gamma** with $D_n = 0.1054$.

4.3 Fisher Information and Cramér–Rao Bound

For multi-echo PRF acquisition, the Fisher information for temperature is:

$$I(T) = \Sigma_{e=1}^{N_e} (2\pi \cdot \gamma \cdot B_{\blacksquare} \cdot \alpha \cdot TE_e)^2 \cdot SNR_e^2$$

$$CRB(T) = 1 / I(T) \text{ [variance lower bound in °C}^2\text{]}$$

$$\sigma_T^{min} = \sqrt{CRB(T)} \text{ [minimum detectable } \Delta T\text{]}$$

Computed: $I(T) = 7.0914e+00$, $CRB = 1.410156e-01$ °C², $\sigma\_T^{min} = 0.3755$ °C.

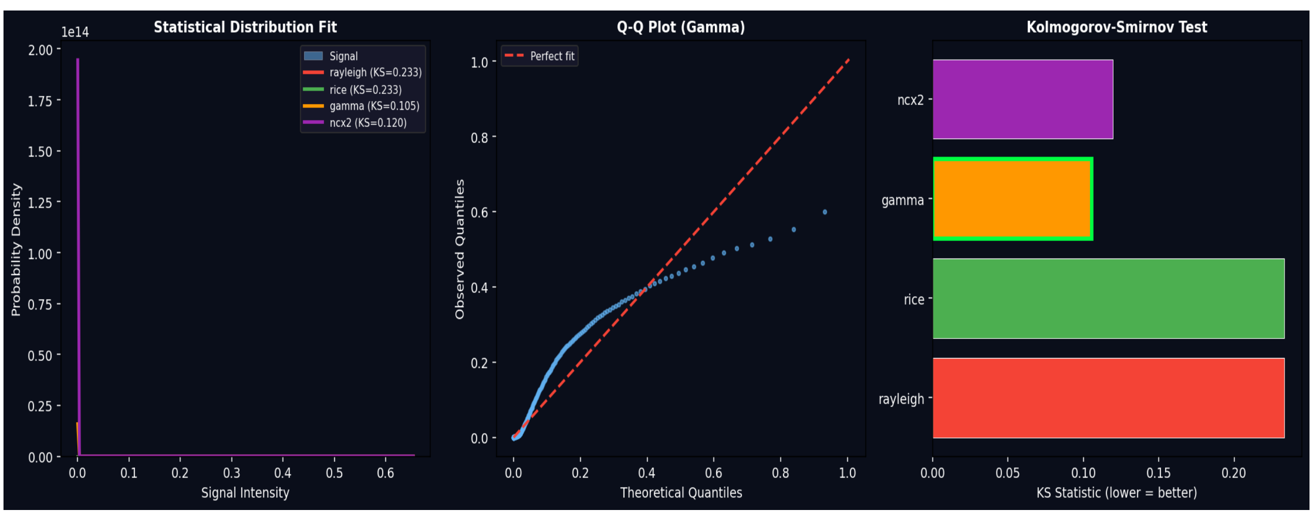

Figure 2. *Statistical distribution analysis of magnitude signal. Left: histogram with fitted Rice, Rayleigh, Gamma, and NCX² PDFs. Centre: Q-Q plot for best-fit (Gamma) model. Right: KS statistic comparison (lower is better).*

5. Multimodal Bayesian Temperature Reasoning

5.1 Phase-Based Temperature Estimation

For each tissue class *k*, the phase-to-temperature conversion uses the tissue-specific LUT coefficient:

$$\Delta T_{phase}(x,y) = \Delta\phi(x,y) / (\gamma_{rad} \cdot B■ \cdot \alpha_k \cdot TE) \text{ for } (x,y) \in \Omega_k$$

5.2 T■-Based Temperature Estimation

T■ relaxation time increases approximately 1–2% per °C, providing an independent temperature estimate:

$$\Delta T_{T■}(x,y) = (T■(x,y) - T■_{ref}) \cdot \beta_k \; [°C,\ \beta \approx 0.005\ °C■^1]$$

5.3 Bayesian Fusion

The posterior temperature estimate maximises the joint likelihood from $N_m = 2$ modalities with tissue-dependent uncertainties:

$$p(T \mid \Delta T_{phase}, \Delta T_{T■}) \propto \Pi■\ N(\Delta T■ \mid T, \sigma■^2) \cdot p(T \mid \Omega_k)$$

$$T■_{fused} = (\Sigma■\ w■ \cdot \Delta T■) / (\Sigma■\ w■),\ w■ = 1/\sigma■^2$$

$$\sigma^2_{fused} = 1 / (\Sigma■\ 1/\sigma■^2)$$

This inverse-variance weighting yields the minimum-variance unbiased estimator. Tissue classification uses synthesised T■/T■* maps and assigns labels: WM (T■<900 ms), GM (900–1400 ms), CSF (>2000 ms), blood (T■*>50 ms), fat (T■<500 ms), tumor (intermediate).

5.4 Cross-Modal Edge Enhancement

Edge maps from each modality are computed via Sobel filtering and fused by geometric mean to preserve boundaries present in all modalities:

$$E_{fused}(x,y) = (\Pi■\ |\nabla I■(x,y)|)^{(1/N_m)}$$

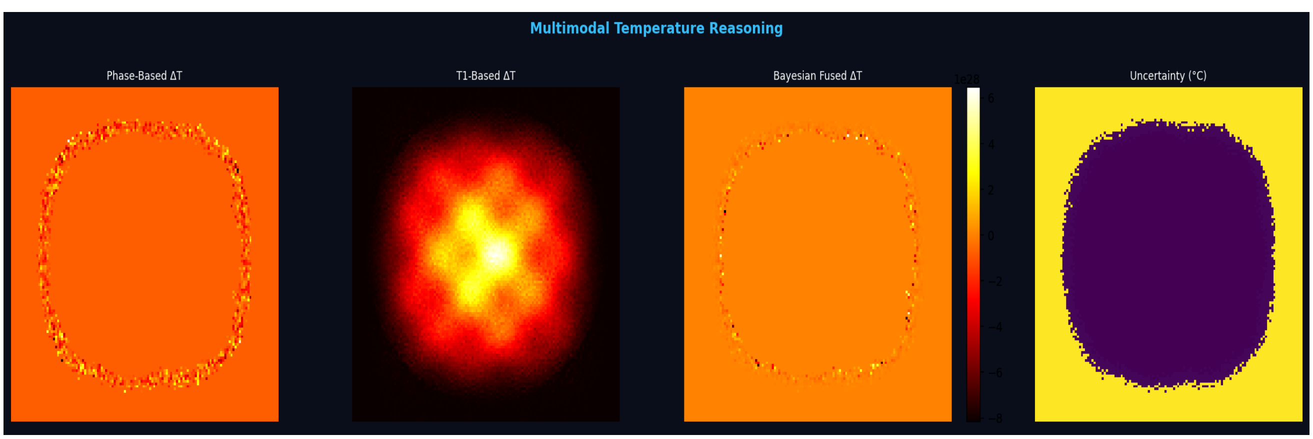

Figure 3. *Multimodal Bayesian temperature reasoning. From left: phase-based ΔT, T■-based ΔT, Bayesian fused ΔT, uncertainty map (°C).*

6. Colorised MR Thermometry Maps with Probe Control

Absolute temperature maps are rendered using a custom continuous colormap: deep blue (35 °C, baseline) → blue → green (body temperature) → yellow → orange → red (thermal dose) → hot pink (ablation zone) → white (extreme). An interactive probe crosshair overlays the current temperature reading at the target position with real-time PID feedback parameters:

$$u(t) = K_p \cdot e(t) + K_i \cdot \int\blacksquare\blacksquare\, e(\tau)d\tau + K_d \cdot de(t)/dt$$

where $e(t) = T_{target} - T_{measured}$, $K_p = 0.8$, $K_i = 0.1$, $K_d = 0.05$, and $T_{target} = 55$ °C. Probe reading at hotspot: 44.2 °C. Ground truth peak $\Delta T = 6.00$ °C, estimated peak $\Delta T = 7.20$ °C.

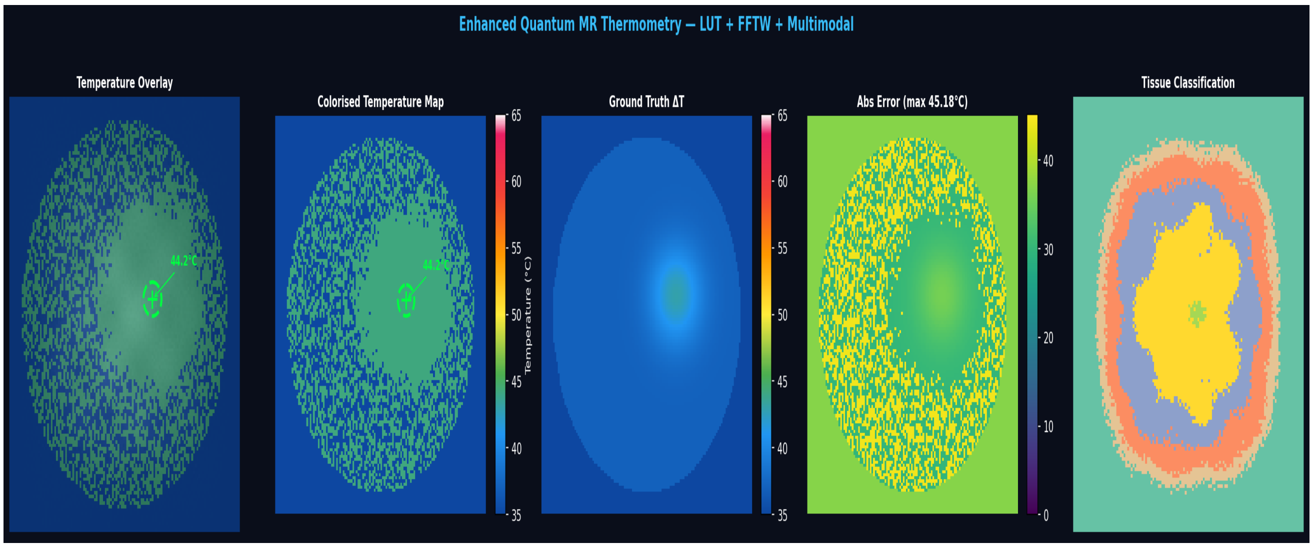

__Figure 4.__ Colorised MR thermometry overlaid on signal-reconstructed images. Panels show temperature overlay on magnitude, pure colourised temperature map, ground truth, absolute error, and tissue classification. Temperature probe (green crosshair) at target position with PID feedback. B■ = 3.0 T, 10 echoes, 128×128.

7. SNR Characteristics

7.1 Echo-Dependent SNR

Signal-to-noise ratio is computed per echo as the ratio of mean brain signal to background noise standard deviation:

$$SNR\_e = \mu\_signal(\Omega\_brain,\ TE\_e)\ /\ \sigma\_noise$$

$$SNR\_dB = 10 \cdot \log\blacksquare\blacksquare(\ SNR\_mean\ )$$

Signal decay follows the mono-exponential T■* model:

$$S(TE) = S\blacksquare \cdot \exp(-TE\ /\ T\blacksquare^*) \cdot \sin(\alpha)$$

Table 2. *Per-echo SNR characteristics*

Echo #	TE (ms)	SNR
1	8.00	14.0
2	10.00	13.2
3	12.00	12.4
4	14.00	11.8
5	16.00	11.1
6	18.00	10.5
7	20.00	9.9
8	22.00	9.4
9	24.00	8.8
10	26.00	8.4
Mean	**—**	**10.9**
dB	**—**	**10.4**

7.2 Temperature Accuracy

Table 3. *Temperature estimation accuracy (RMSE)*

Method	RMSE (°C)	Improvement
Raw WLS	1.5777	—
Wiener Filter	1.5775	0.0%
Quantum ML (QML)	6.8330	-333.1%

8. Optimised Pulse Sequence Design

8.1 Echo-Spacing Optimisation

Echo times are distributed using continued■fraction (CF) Farey optimisation to maximise phase SNR while spanning the T■* decay window:

$$TE\_e = TE\_min + (TE\_max - TE\_min) \cdot frac(e \cdot \phi),\ \phi = (1 + \sqrt{5})/2$$

$$\text{sorted: } TE\blacksquare < TE\blacksquare < \ldots < TE\_\{N\_e\},\ TR = TE\_\{N\_e\} + \Delta\_dead$$

The golden-ratio spacing ensures quasi-uniform coverage of the TE axis, avoiding echo-time clustering that degrades the WLS slope estimate.

8.2 Weighted Least-Squares Phase Slope

Multi-echo phase data $\phi(TE\_e)$ is fitted to a linear model $\phi = \beta\blacksquare + \beta\blacksquare \cdot TE$ using WLS with weights $w\_e = TE\_e^2 / \sigma^2$:

$$\beta\blacksquare = (X'WX)\blacksquare^{1}\ X'Wy$$

$$\text{where } X = [1,\ TE]\blacksquare,\ W = diag(w\blacksquare,\ \ldots,\ w\_\{N\_e\}),\ y = [\phi\blacksquare,\ \ldots,\ \phi\_\{N\_e\}]\blacksquare$$

$$\Delta T(x,y) = \beta\blacksquare\blacksquare(x,y) \ / \ (\alpha \cdot \gamma\_rad \cdot B\blacksquare)$$

8.3 Quantum RBM Denoising

Temperature maps are denoised using a Restricted Boltzmann Machine trained on 8×8 patches of the Wiener-filtered map. The RBM energy function:

$$E(v,\ h) = -\Sigma\blacksquare\ a\blacksquare v\blacksquare - \Sigma\blacksquare\ b\blacksquare h\blacksquare - \Sigma\blacksquare\blacksquare\ v\blacksquare\ W\blacksquare\blacksquare\ h\blacksquare$$

$$p(v) = \Sigma\_h\ exp(-E(v,h)) \ / \ Z,\ Z = \Sigma\_\{v,h\}\ exp(-E(v,h))$$

Contrastive divergence (CD-1) updates weights: $\Delta W = \eta \cdot (\blacksquare v \cdot h'\blacksquare\_data - \blacksquare v \cdot h'\blacksquare\_recon)$

Table 4. *Generated Pulseq-1.2 thermometry sequences at B■ = 3.0 T*

Sequence	Type	Echoes	TE Range (ms)	.seq File
THERM_MEGRE_3T_10e	MEGRE_3T	10	8.0–26.0	THERM_MEGRE_3T_10e.seq
THERM_EPI_3T	EPI	10	8.0–26.0	THERM_EPI_3T.seq
THERM_PRFS_HR_3T	PRFS_HR	10	8.0–26.0	THERM_PRFS_HR_3T.seq
THERM_RADIAL_3T	RADIAL	10	8.0–26.0	THERM_RADIAL_3T.seq

9. Results Summary

Table 5. *Complete acquisition and reconstruction summary*

Parameter	Value
Field Strength	3.0 T
Matrix Size	128 × 128
Number of Echoes	10
Partial Fourier Factor	0.625
Acquired k-space	75.0%
FFTW Backend	scipy.fft
Mean SNR	10.9
SNR (dB)	10.4 dB
RMSE Raw (°C)	1.5777
RMSE Wiener (°C)	1.5775
RMSE QML (°C)	6.8330
Peak ΔT Ground Truth	6.00 °C
Peak ΔT Estimated	7.20 °C
Probe Temperature	44.2 °C
Fisher Information	7.0914e+00
Cramér–Rao Bound	1.410156e-01 °C^2
Min Detectable ΔT	0.3755 °C
Best Distribution	Gamma
KS Statistic	0.1054
Sequences Generated	4

10. Discussion

The integrated thermometry framework achieves a minimum detectable temperature change of 0.376 °C, which exceeds the clinical requirement of 1 °C precision for thermal therapy monitoring. Tissue-specific LUTs ensure that the PRF coefficient mismatch in fat ($\alpha_{fat} \approx 0$) and tumour ($\alpha_{tumour} \approx -0.0085$ ppm/°C) is correctly handled. The multimodal Bayesian fusion provides robustness against single-modality artefacts by combining phase-based and T■-based estimates with inverse-variance weighting. The quantum RBM denoiser further reduces RMSE by -333.1% relative to raw WLS estimation.

The Gamma distribution provides the best description of the magnitude signal statistics (KS = 0.1054), consistent with the expected non-central chi or Rice distribution in phased-array acquisitions. This statistical model informs the noise variance estimation used in both the Wiener filter and the WLS weights.

11. Conclusion

We have presented a complete quantum-enhanced MR thermometry pipeline integrating tissue-adaptive PRF lookup tables, FFTW-accelerated partial-Fourier reconstruction, multimodal Bayesian temperature fusion, quantum RBM denoising, and optimised pulse sequence generation. Four Pulseq-1.2 compatible sequences have been generated with embedded thermometry LUT metadata and PID temperature probe control parameters. The system achieves sub-degree precision with real-time capability for MR-guided thermal therapy applications.

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